A System for Burned Area Detection on Multispectral Imagery

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***Abstract*— The current remote sensing (RS) open data policy for multispectral (MS) missions such as Sentinel-2 and Landsat-8, together with the availability of free cloud distributed processing platforms such as Google Earth Engine, makes it possible the quick generation of burned area (BA) products even for nonexperts in the field. Indeed, fires and BAs can be detected using burn severity indices, which are usually obtained by simple band algebra operations. However, simple approaches can aid BA estimation only if typical error patterns are known and accounted for, especially when working at large (e.g., continental) scales. This article proposes an automatic BA detection system based on burn severity index thresholding, which integrates dedicated false and missed alarm mitigation strategies to improve the detection accuracy. The system is tested on Sentinel-2 and Landsat-8 data over ten different locations in Europe and spanning year 2018. Three known burn severity indices plus a custom one defined to improve the performance in the considered study area are under study. Results show that burned index thresholding is possible within accuracy bounds slightly larger than the state of the art, which is acceptable by considering the proposed simplified processing framework.**

***Index Terms*— Burn severity index, burned area (BA), Landsat-8, multispectral (MS) images, Sentinel-2.**

I. Introduction

FIRE events (burning biomass) affect, and in some cases are affected by, several key environmental processes, such as vegetation regrowth, land cover change, desertification, hydrological cycles, and, above all, atmospheric emissions [1], [2]. Because of its important socioeconomical and environmental implications, the fire disturbance has been recognized by the Global Climate Observing System (GCOS) [3] as one of the essential climate variables (ECVs), being burned area (BA) its primary descriptive variable [4]. Differently from active fires (AFs), which are from the remote sensing (RS) perspective instantaneous phenomena that can be identified through spectral anomalies in thermal bands, BAs have a relatively longer persistent nature [5]. Indeed, they can be identified by charcoal residue and vegetation scar, having both quite specific and lasting spectral signatures [6]. However, the variability of these signals is often strong and increases with time from the event. The dark spectral signal due to charcoal deposition is relatively short-lived as factors such as wind scattering and rain, in a period of a few weeks to some months after the fire, can attenuate it significantly. The vegetation scar signal is more persistent whereas less specific, as partial or total vegetation removal may result from other practices such as harvesting, grazing, or windthrow. Moreover, even stronger signal mitigation is often given by vegetation regrowth and the underlying soil signature [7]. It follows that the key for the successful mapping of BAs is appropriate spectral, spatial, and temporal characteristics of satellite-borne RS optical missions [8].

The open data policy applied by multispectral (MS) satellite missions, such as Sentinel-2 and Landsat together with the availability of planetary-scale data analysis cloud-distributed web-browser powered platforms such a Google Earth Engine (GEE), made it possible in the very recent years, both for experts and untrained practitioners, the prompt generation of thematic products at large scales due to an unprecedented real-time image processing capability [9]. This implies the possibility to generate quickly and at no cost useful reference maps and datasets, and this is especially true for BA mapping being burned vegetation often identifiable through relatively simple spectral index analysis [10]–[12]. On the other hand, simple and intuitive spectral methods, such as burned index thresholding, must be applied consciously and awareness is required on the user regarding typical error patterns and possible mitigation strategies that can be applied to improve the accuracy of the results, especially when working at large scales [6], [13].

In this article, a multitemporal system is proposed to perform BA detection via spectral index analysis on Sentinel-2 and Landsat-8 MS images. To overcome well-known limitations of spectral-based methods such as false detections for certain types of land cover and missing detection of less severe burned spots, some mitigation techniques are proposed. For

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TABLE I
SUMMARY OF MULTISPECTRAL SENSORS USED IN BA PRODUCTS GENERATION

Instrument	bands	resolution [m]	revisit [days]
SENTINEL-2	13	10, 20, 60	2 to 5
LANDSAT-8	8	30	16
MODIS	36	250, 500, 1000	1
MERIS	15	300	3
AVHRR	4-5	1000	1
SPOT-VGT	4	1150	1

ease of application, the BA detection algorithm is described in this work for homogeneous data (same source). It is anyway possible to extend its philosophy to more than one source of MS imagery and/or SAR in a more fusion setting, but this is beyond the scope of this work. The algorithm is tested over the so-called European Forest Fire Information System (EFFIS) area ([14], indicating extended Europe, therefore including a variety of ecosystems and land cover), where issues related to temporal resolution (5 and 16 days for Sentinel-2 and Landsat-8, respectively), such as cloud coverage and vegetation regrowth, are expected to be less severe than in other parts of the world such as the Tropical belt or Siberia.

This article is structured as follows. In Section II, an overview on the state-of-the-art of BA mapping with large-scale application is first given, followed by a presentation of the rationale of this work. Section III presents the proposed system for BA detection on high-resolution images in a general setting. Specific implementation of the system for Sentinel-2 and Landsat-8 data is described in Section IV. Section V presents the experimental data and results, including dataset presentation, implementation, reference data, and evaluation of the performance. Conclusions are drawn in Section VI.

II. BA PRODUCTS AND ALGORITHMS

The effectiveness of BA detection from space strongly depends on spectral/spatial resolution and observation frequency. Therefore, MS mission acquisition plans are particularly useful in proportion to their ability to provide immediate, or nearly immediate, postfire acquisitions. However, as shown in Table I, which shows a list of MS sensors used in BA detection, increasing spatial/spectral detail always corresponds to decreasing temporal resolution. This basic tradeoff is the main challenge in BA detection, which has been grounded for years in the high spectral and temporal detail of the Moderate Resolution Imaging Spectroradiometer (MODIS), even at the cost of low spatial resolution, and which is now witnessing a new era due to the unprecedented characteristics of Sentinel-2 data. This section first provides a review of the main algorithms and products obtained so far by emphasizing the path from moderate to high resolution in BA large-scale applications. This analysis is then followed by a statement of the main strengths of high spatial resolution data from which stems the rationale of this work.

A. Moderate Spatial Resolution Products

The MODIS Standard Fire products provide an important contribution to the NASA Land Cover and Land Use Change (LCLUC) program and the international Global Observation of Forest Cover (GOFC) project. The latest version (Collection 6, superseding the heritage Collection 5.1) of the MODIS Global Burned Area Product was released in 2017 [15], consisting of two global BA time series (from year 2000 to the present, at 500 m) named MCD45A1 and MCD64A1. The MCD45A1 was derived from a change detection algorithm based on a bidirectional reflectance model [16], whereas the MCD64A1 is derived from a hybrid algorithm that uses both the reflectance changes and the thermal anomalies associated with biomass burning [17], [18] and represents the current standard of MODIS BA products. The algorithm proceeds through multiple stages, each applied to a single MODIS tile. The main task is to produce composite imagery summarizing persistent changes in the time series of a burn-sensitive vegetation index and then use spatial and temporal AF information to estimate the statistical models of burn- and unburn-related changes. Then, probabilistic thresholds are computed for the classification of individual 500-m grid cells as either burned or unburned [15]. MERIS images have been used for generating global BA products as well, with shorter time series than VGT and MODIS (anyway limited to the period of available MERIS input data, i.e., 2005–2011), but with higher spatial resolution (300 m) [19], [20]. This product was named FireCCI41, being itself a precursor of the ultimate Fire product from ESA's Climate Change Initiative (CCI) Programme (cf. Section II-C).

B. Landsat Products

Capitalizing on the longest high-resolution MS time series ever available, the production concerned with the Landsat Burned Area Essential Climate Variable (BAECV), developed by the U.S. Geological Survey (USGS), returned a map of BAs all across the conterminous United States (CONUS) in the time period 1984–2015 [21], [22]. The BAECV algorithm [21] is a supervised approach able to generate burn probability surfaces via boosted regression of several variables, such as spectral bands and indices from individual Landsat scenes, lagged reference conditions, and change metrics between the scene and reference predictors, including the normalized difference vegetation index (NDVI), and many other vegetation indices, including normalized burn ratio (NBR) and NBR2 (an alternative version of NBR). The burn probabilities are then composited to per-pixel maximum annual burn probabilities, followed by a thresholding and segmentation process to generate binary annual burn classification images. For the production phase, monitoring trends in burn severity (MTBS) data [23] over observed BAs have been used both for training and evaluation [22].

C. CCI Fire Products

The CCI Fire project, which is part of the ESA's CCI program [24], aims to develop long-term time series of BA

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products adapted to the needs of climate modelers [20]. Its focus is also on mapping BAs using European sensors such as the along-track scanning radiometer (ATSR), SPOT-VGT, and MERIS, and in comparing the performance of the results with other existing datasets. Throughout the life of CCI Fire project, joint efforts with other projects lead to the recently published Burned Area Reference Dataset (BARD), a valuable source for validation of global products mainly obtained via photointerpretation of already existing datasets [25].

The last CCI Fire product version FireCCI50 is a global BA product obtained from a full time series 2001–2016 of Terra-MODIS Red (R), near-infrared (NIR), and thermal anomaly data at 250-m resolution. For the production [26], daily images were aggregated into monthly composites to select the pixels most clearly associated with a burned signal. Then, the BA algorithm follows a two-phase approach, detecting in the first phase the most clearly burned pixels (named seeds) as local minima of NIR through a moving window and improving the burn shape detection in the second phase using a growing procedure around the seed pixels. In the growing stage, an additional threshold based on the cumulative density function of the Global Environmental Monitoring Index (GEMI) is used. The first and second phases are targeted to reduce commission error (CE) and omission error (OE), respectively.

In addition to global BA products, the CCI Fire project is generating a fire database for the African continent based on high-resolution sensors on board the Sentinel-2 and Sentinel-1 satellites [13]. The work includes a spectral analysis for the identification of the most informative spectral bands and indices useful for BA detection. The rationale of the algorithm is similar to that used for obtaining MODIS BA products, i.e., an initial BA map is created in the first step, from which tile-dependent statistics are extracted for the second step. In the first step, BA probability is computed over two consecutive Level-2A Sentinel-2 image products by means of mid-infrared burned index (MIRBI) and NBR2 spectral indices, and the NIR band. The initial choice of highly likely BAs is obtained through fixed thresholds on the probability values. BAs confirmed by intersection with AF hotspots (MCD14ML) detected in between the two dates of acquisition are lately used to obtain tile-dependent statistical thresholds for each predictive variable in a two-phase burned strategy, following a modified method from [27].

D. Discussion on High Spatial Resolution

The advent of the Sentinel-2 mission has driven a change of paradigm in the way products can be generated because of its unprecedented characteristics. It spans 13 spectral bands at 10 m in the visible and near-infrared ranges, 20 m for vegetation red-edge and shortwave infrareds, and 60 m in spectral ranges sensitive to water vapor, cirrus, and aerosols. Moreover, it guarantees almost worldwide coverage of lands with a revisit time of five days at most (at the equator), reducing to 2–3 days at midlatitudes. For more than two decades, MODIS has provided the most frequent and coherent source of data for BA detection and mapping (see Section II), having as major strengths a remarkable number of bands available (36) and a revisit time of one day. However, several studies demonstrated that only specific spectral bands/indices are essential for the purpose of BA detection [13], [28]. More specifically, band selection can be restricted to the NIR and the long and short portions of the short-wave infrared (SWIR), useful for the calculation of NBR, NBR2, and MIRBI indices [29]. Therefore, in comparison with MODIS data, Sentinel-2 imagery can provide quality information for BA detection at the significantly higher resolution of 20 m but at the expense of a little increased revisit time (2–5 days), which is comparable anyway to that of MERIS (see Table I). From an operational point of view, Landsat data are more critical. Although Landsat is the longest series of MS data ever recorded, compared to Sentinel-2, it shows limitations due to temporal sampling. Indeed, the sparse resolution of 15 days, jointly considered with cloud cover, negatively impacts the capability of detecting postfire strong BA signal, thus resulting in potential high rate of missed detections.

As previously mentioned, one of the acknowledged main limitations of current BA algorithms is the typical rate of OE [30], which is still critically high on MODIS products (rate 60%) [18]. This is reduced by more than a half on Sentinel-2 products over the African continent (rate 26.5%) [13] and by one third over Italy (rate 40%) [31]. Comparative analyses traced the missed detections issue back to two main causes. The first one relates to spatial resolution. Significant differences between global MODIS products and continental/subcontinental high-resolution ones were identified mainly over regions where small fires ($<$100 ha, i.e., four MODIS 500-m sized pixels) accounted for a relevant part of the actual BA, as they were not detected because of the coarse pixel size (only using Satellite-A products) [13]. The second one comes from the common practice of relying on MODIS AF hotspots (even if properly preprocessed [13], [18]) for estimation of local-dependent characteristics of the BA spectral properties. In fact, all specific fires that are systematically missed by MODIS thermal anomalies do not contribute to fire description, thus resulting in possibly significant underestimation of BA. It is worth noting that all shortcomings applied to reduce OEs can only grow-up and refine already detected burned patches and are unable to completely recover missed fires [13]. A similar argument applies also to BA detection methods based on training-classification approaches [22]. Here, in fact, there is no guarantee for the sampling strategy to cover all fire types, thus resulting in a potential under-representation of classes in the feature space. Not to mention that, collecting labeled samples is always a costly task, often unfeasible in operational contexts. Various product assessment activities carried out in the projects mentioned before pointed out, almost unanimously, that CE is intrinsically related to difficult spectral discriminability between BAs and other specific land cover changes associated mainly with agricultural practices on croplands that take place especially during harvesting and planting seasons [13], [21], [30]. Other observed sources of error are related to inaccurate masking of clouds and several other types of nonburnable objects. Of course, the quality of BA products increases with the quality of Level-2A associated products. This is proven, for example, by the positive

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trend characterizing subsequent versions of CCI Fire-related products [13], [26].

E. Rationale of the Work

The aim of this work is to address the generation of multitemporal BA maps by only using high-resolution MS imagery, being Sentinel-2 and Landsat-8 the main targets. Given the abovementioned considerations, the proposed methodology leverages on the following aspects.

1) *Spectral Detail in NIR and SWIR:* To maximize BA spectral representation on different scenarios and conditions such as vegetation types and different kinds of fire events such as wildfires or human-ignited fires, and land use practices. We account that Sentinel-2 and Landsat-8 provide similar spectral bands in the portion of the spectrum relevant for BA detection (cf. Table III).
2) *Bitemporal Approach:* From the RS viewpoint, the fire event can be seen as an abrupt change. Therefore, essential to BA detection is quality (cloud-free, nonsaturated, intraseasonal) preevent and postevent measurement of spectral response from the observed area.
3) *Independence From Low-Resolution Data:* The strong relationship linking contemporary algorithms to MODIS AF hotspots may represent a constraint on the attempt of reducing certain types of OEs.
4) *Temporal Resolution:* Needed for increasing probability of immediate postfire acquisition, therefore reducing OE rates. This is a drawback of Landsat-8, which has higher revisit time if compared to Sentinel-2 (cf. Table I).

III. Multitemporal BA Detection System

As shown in Fig. 1, the proposed multitemporal system for BA detection is made of two components: 1) a multitemporal image composer and 2) a BA detector. The first module is deputed to the generation of a bitemporal image pair where the "post" image is the target image, whereas the "pre" image is a multitemporal composite of the most recent nonoccluded (clouds, shadows, and so on) pixel values available in a fixed period of time before the target date. The second module performs binary change detection on the composite image pair based on spectral index analysis to detect BAs. In this section, the system is described in general terms, whereas specific implementation for Sentinel-2 and Landsat-8 data is described in Section IV.

A. Notation

The components of an MS image are notated as $X_b^{L,t}$, where b specifies the spectral band, t is the time of acquisition, and L is an identifier for the spatial location (a tile in a particular world reference gridded system, e.g., an MGRS tile for Sentinel-2 or a path-row pair for Landsat-8), which is later referred to as spatial frame for brevity. Pixel coordinates are notated within parenthesis as $X_b^{L,t}(x, y)$, where x and y are row and column subscripts, respectively.

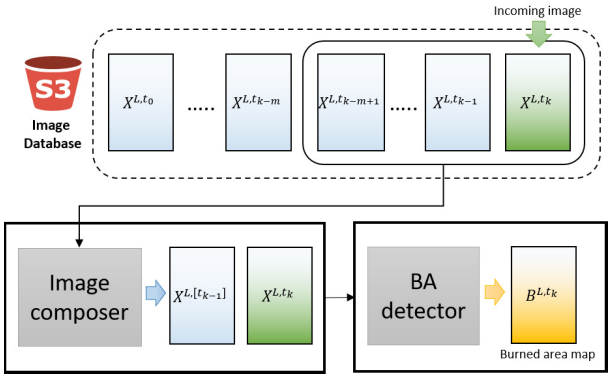


Fig. 1. High-level scheme of the BA detection system. For a target image X^{L,t_k}, the system detects BAs at date t_k using a time series of m images.

B. Assets

Mandatory requirements for the input imagery are: 1) surface reflectance (SR) data for the computation of spectral indices based on atmospherically corrected data and 2) availability of thematic products associated with each image for the identification of burnable (or burned) and nonburnable land cover (e.g., the Landsat CFMask or Sentinel-2 Scene Classification products). We therefore assume to have for each image $X^{L,t}$ a mask product $X^{L,t}_{\text{mask}}$ taking values 0 or 1 on pixel (x, y) if this is classified as nonburnable or burnable, respectively. Nonburnable objects are, for instance, clouds, shadows, water, snow, ice, and saturated pixels.

C. Bitemporal Image Composer

The BA detection system targets the specific date t_k and triggers the process to map pixels that are likely to be burned between dates t_{k-1} and t_k. To recover information over occluded pixels at date t_{k-1}, earlier dates are used. In order to limit spectral variability due to possible temporal decorrelation we impose, the analysis is performed within a limited temporal interval $[t_k - \Delta t, t_k]$. The extracted m images from the base time series are then $X^{L,t_{k-m+1}}, \ldots, X^{L,t_{k-1}}, X^{L,t_k}$ and these are used by the system for the detection at date t_k. For later convenience, it is useful to define for each pixel (x, y) the set of dates between t_{k-m+1} and t_{k-1} that correspond to nonoccluded (burnable) values

$$T_{x,y}^{L,t_k} := \left\{ t \in \{t_{k-m+1}, \ldots, t_{k-1}\} : X^{L,t}_{\text{mask}}(x, y) = 1 \right\} \quad (1)$$

as this will be used as an index set for computations spanning the temporal dimension and involving only meaningful spectral values. The composite "pre" image, denoted as $X^{L,[t_{k-1}]}$, is obtained through a multitemporal composition of the the most recent nonoccluded pixels between dates t_{k-m+1} and t_{k-1}. Thus, we have

$$X_b^{L,[t_{k-1}]}(x, y) := X_b^{L,t^*_{x,y}}(x, y) \quad (2)$$

where $t^*_{x,y} := \max\{t : t \in T_{x,y}^{L,t_k}\}$.

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D. BA Detection

The BA detection phase divides into three steps:

1) candidate burned pixels detection;
2) candidate burned pixels screening (seeds);
3) BAs full reconstruction.

1) Candidate Burned Pixels Detection: As previously discussed, spectral burn severity indices carry important information for inferring the presence of BAs. Bitemporal indices (e.g., dNBR) are known to perform better than single date indices (e.g., NBR) on this task [32]. High values of bitemporal burn severity indices usually indicate for likely presence of burned matter. We therefore assume, as it is done in most of the literature, that it is possible to discriminate between burned and unburned pixels according to thresholds set on the burned severity index values [13], [21], [30]. Let us denote by X_{bsi}^{L,t_k} the burn severity index chosen for the analysis on the target date. To keep generality, we just assume for the system definition that this index can be obtained via a functional relationship of the type

$$X_{\text{bsi}}^{L,t_k} := f_{\text{bsi}}\left(X^{L,[t_{k-1}]}, X^{L,t_k}\right). \tag{3}$$

Let us denote by ℓ^{L,t_k} the threshold value suitable to the discrimination of burned pixels in the spatial frame L at the date t_k. This value defines the map of candidate burned pixels X_c^{L,t_k} according to the rule

$$X_c^{L,t_k}(x,y) = \begin{cases} 1, & X_{\text{bsi}}^{L,t_k}(x,y) \geq \ell^{L,t_k} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

for every pixel (x, y) in the spatial frame L.

It is acknowledged in the literature that the bitemporal spectral signature observed on BAs can be matched over other types of change as well. This might happen on noisy pixels and more generally over specific cultivated areas [13], [21], [30]. Therefore, we characterize two main categories of false detections: sparse detections showing up mainly as salt-and-pepper noise, and more coherent and clustered detections (mainly related to crops). These two categories of errors must be handled through different approaches. The first category is addressed in the pixel screening step via a pixel-based statistical test, whereas the second one is addressed in the reconstruction step following a more contextual approach.

2) Candidate Burned Pixels Screening: Based on the main assumption that burned matter implies an abrupt change in the spectral multitemporal domain statistically behaving like an anomaly if compared to other possible changes, the system includes a step for the discrimination between strongly and weakly detected changed pixels in such a way the weak ones (the potential source of false alarms) can be eliminated from further analysis. This weakness is measured in terms of statistical accordance of a pixel spectral value as measured on the target date t_k and all its previous instances from t_{k-m+1} to t_{k-1}. Therefore, after choosing a suitable spectral index $X_{\text{siss}}^{L,t}$ (spectral index for statistical-based screening), the full time series of nonoccluded values is analyzed as follows. For any candidate pixel (x, y), the spectral value $X_{\text{siss}}^{L,t_k}(x,y)$ is tested against its normality based on parameter estimates made with all nonoccluded spectral values measured between t_{k-m+1} and t_{k-1}. The test rejects the hypothesis that the sample is normal with parameters estimated from the past observations w.r.t. the target date (therefore a false alarm) if its value has a normalized distance from the mean that is greater than a fixed tolerance α. Mathematically, given mean and variance

$$\mu_{x,y}^{L,t_k} = \sum_{t \in T_{x,y}^{L,t_k}} \frac{X_{\text{siss}}^{L,t_k}(x,y)}{\#\left(T_{x,y}^{L,t_k}\right)} \tag{5}$$

$$\sigma_{x,y}^{L,t_k} = \sum_{t \in T_{x,y}^{L,t_k}} \frac{\left[X_{\text{siss}}^{L,t_k}(x,y) - \mu_{x,y}^{L,t_k}\right]^2}{\#\left(T_{x,y}^{L,t_k}\right)} \tag{6}$$

any candidate pixel (x, y) is confirmed as burned if

$$X_{\text{siss}}^{L,t_k}(x,y) > \mu_{x,y}^{L,t_k} + \alpha * \sigma_{x,y}^{L,t_k}. \tag{7}$$

Screened pixels are in the first stage those for which $X_c^{L,t_k}(x,y) = 1$ (i.e., candidates) that also verify (7).

A second filtering stage is necessary for the screening step to deal with the residual errors due to partial erosion of false alarm patches eliminated in the previous step. To this aim, a morphological filter is applied to remove small objects from the screened pixels. Having fixed p the number of pixels, connected components with size greater than p are those passing the morphological area filter. Smaller objects are considered as noise or residuals of erroneous detections. The parameter p must be properly chosen by considering the pixel size and other characteristics of the MS sensor. Let us denote by X_s^{L,t_k} the map representing pixels that passed the screening step, i.e., those pixels showing the strongest signal from the BAs. The map X_s^{L,t_k} can be viewed as a collection of seeds from which the full burned patches can be reconstructed.

3) BAs Full Reconstruction: The reconstruction step aims to include less severely burned pixels spatially adjacent to the seeds according to an inclusion principle. The whole process is controlled by a constant that decides whether the inclusion is reliable or not and this is done at the level of connected components. First, an abundant detection map X_{ab}^{L,t_k} is obtained similarly as the candidate one in (4) but using a lower threshold h (thus including more pixels). The term abundant refers to the fact that the threshold h is chosen so that it is smaller than any of the local thresholds ℓ^{L,t_k} to ensure that the enlarged regions always contain the seeds. Then, for each connected component C in X_{ab}^{L,t_k}, its growing ratio g_C is computed as the area of the seeds included in it over the area of the region itself

$$g_C = \frac{\#\left(C \cap X_s^{L,t_k}\right)}{\#(C)}. \tag{8}$$

By definition, we have $0 < g_C \leq 1$. Within the perimeter of a recently occurred fire event, it is expected that the area of high severity has a relative proportion with respect to the whole BA and this proportion can be bounded to some constant from below. Too small values of this ratio are indeed likely indicating great unbalance, possibly representative of something different than a burning event. By this reasoning, the current connected component C is successfully considered as the full reconstruction of its seeds if $g_C > i_R$, where i_R is a

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TABLE II
VARIABLES TO BE ASSIGNED FOR A SPECIFIC IMPLEMENTATION OF THE BA DETECTION SYSTEM

Type	Var.	Description
System	$X^{L,t}$	image data over spatial frame L at time t
	X_{mask}	scene classification mask
	Δt	time span of the time series
Spectral	X_{bsi}	burn severity index
	$\ell^{L,t}$	bsi threshold for candidate burned pixels detection
	X_{ssis}	spectral index for statistical-based screening
	h	bsi abundant threshold for burned area full reconstruction
	α	normalized distance to the mean
Morphological	p	min size (pixels) of burned area
	i_R	min inclusion ratio for burned area full reconstruction

fixed inclusion ratio threshold. On the contrary, the clustered seeds within an unbalanced region $g_C < i_R$ are rejected being considered as false detections. The collection of successfully reconstructed BA patches is the BA map denoted by B^{L,t_k}, which is the final output of the BA detection module of the system.

IV. SYSTEM IMPLEMENTATION FOR SENTINEL-2 AND LANDSAT-8 DATA

Implementing the proposed system for specific data requires assigning values to a set of variables, including system data and parameters, spectral indices and thresholds, and morphological parameters. For convenience, all of them are summarized in Table II. In the remainder of this section, the rationale for defining these variables to tailor the system on Sentinel-2 and Landsat-8 data is presented. Experimental results presented in Section V refer to the setting presented here.

A. System Data and Parameters

1) Image $X^{L,t}$ and Scene Classification X_{mask} Data: Sentinel-2 and Landsat-8 SR data are freely available within Level-2 processing-level products generated using the sen2cor[1] processor and the land surface reflectance code (LaSRC),[2] respectively. Scene classification is an additional product[3] of Sentinel-2 Level-2A, being burnable/burned pixels those classified as UNCLASSIFIED (DN = 7). Landsat-8 scene classification is obtained through CFMASK,[4] where burnable/burned pixels are those classified as DESIGNATED FILL (DN = 0).

[1]https://sentinel.esa.int/web/sentinel/user-guides/sentinel-2-msi/processing-levels/level-2

[2]https://www.usgs.gov/land-resources/nli/landsat/landsat-collection-2-surface-reflectance

[3]https://sentinel.esa.int/web/sentinel/technical-guides/sentinel-2-msi/level-2a/algorithm

[4]https://www.usgs.gov/land-resources/nli/landsat/cfmask-algorithm

2) Temporal Span Δt: The temporal span limitation is a mandatory constraint that prevents the system from analyzing strongly uncorrelated spectral values. Therefore, we impose a (pseudo) intraseasonal time span interval for the inclusion of images within the time series, i.e., $\Delta t = 2$ months.

B. Spectral Indices and Parameters

1) Burn Severity Index X_{bsi}: The key for the identification of BAs is the characterization of postfire effects on vegetation (e.g., exposure of bare soil, presence of charcoal, or ash) both in the spectral and temporal domains. This is typically done using spectral indices aimed at enhancing and underlining these effects in the data. In this article, NBR, NBR2, MIRBI, and a custom *ad hoc* index specifically derived through empirical analysis in the EFFIS area are considered for evaluation.

The typical observation of BA presents low response in the visible range of the spectrum due to the presence of dark matter and stronger signal in the shortwave infrared region due to higher temperatures and the low evapotranspiration [33]. Therefore, NBR is defined as [10]

$$\text{NBR} = \frac{\text{NIR} - \text{SWIR}}{\text{NIR} + \text{SWIR}} \tag{9}$$

and is commonly used in the literature to determine fire severity in proportion to its variation between time instants, as per formula [11]

$$\text{dNBR} = \text{NBR}_{\text{pre}} - \text{NBR}_{\text{post}}. \tag{10}$$

As a matter of fact, there exist no assessment of NBR and/or dNBR automatic and unsupervised binary classification of BA (unburned/burned) in high-resolution MS imagery at large scale. To date, we only know that dNBR typically performs as well as, or better than, NBR in the binary classification task [32]. The NBR2 is a modified version of NBR, which is defined as [34]

$$\text{NBR2} = \frac{\text{SWIR}_1 - \text{SWIR}_2}{\text{SWIR}_1 + \text{SWIR}_2} \tag{11}$$

and it was introduced to highlight water sensitivity in vegetation by exploiting SWIR channels. The same bands are used in MIRBI, which was derived through linear regression as [35]

$$\text{MIRBI} = 10 * \text{SWIR}_2 - 9.8 * \text{SWIR}_1 + 2 \tag{12}$$

with the purpose of enhancing the sensitivity to spectral changes caused by burning by keeping spectral variability relatively low.

Typically, the performance of spectral indices for BA detection is very dependent on many factors, such as radiometric conditions, local climate, and diversity of ecosystems. For our analysis in the EFFIS area, we tried to mitigate for this variation using a particular implementation of a normalized difference index (i.e., a band algebra expression of the type $\text{ndi}(a, b) = (a - b)/(a + b)$ [10], [36]) in which the statistical distributions of unburned and burned classes are better distinguishable. To get this sharpened behavior, we identified the cellulose absorption index (CAI) [37]

$$\text{CAI} = \frac{\text{SWIR}_2}{\text{SWIR}_1} \tag{13}$$

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TABLE III
MULTISPECTRAL BANDS RELEVANT TO BA DETECTION

band	Landsat-8 id	res.	range [nm]	Sentinel-2 id	res.	range [nm]
NIR	B05	30	850-880	B08	10	758-905
$SWIR_1$	B06	30	1570-1650	B11	20	1542-1684
$SWIR_2$	B07	30	2110-2290	B12	20	2072-2312

which indicates exposed surfaces containing dried plant material, as the index with high response on the burned matter, and a quotient version of NBR

$$q\text{NBR} = \frac{\text{NIR}}{\text{SWIR}_2} \tag{14}$$

as the index with low response on the burned matter. The proposed normalized difference BA index, notated NDBI for simplicity, is therefore defined as NDBI := ndi(CAI, qNBR). In Table III, the Sentinel-2 and Landsat-8 bands relevant for spectral index calculations are listed.

Establishing the functional relationship f_{bsi} for the definition of X_{bsi}^{L,t_k}, as described in Section III-D1, just implies bitemporal differencing for NBR, NBR2, and MIRBI. Thus, we have, similar to dNBR [cf. (10)], also dNBR2 and dMIRBI. For NDBI, the temporal dimension is included differently. More specifically, we enforce the binary representation of the unburned/burned attributes by considering both the single date and the differential contributions of NDBI. This is formally obtained by averaging its measurement at the target date t_k with its difference between the target t_k and the (composite) predate $[t_{k-1}]$. Thus, we define the weighted bitemporal NDBI as

$$\begin{aligned} X_{\text{wNDBI}}^{L,t_k} &= \frac{1}{2}\left[X_{\text{NDBI}}^{L,t_k} + \frac{1}{2}\left(X_{\text{NDBI}}^{L,t_k} - X_{\text{NDBI}}^{L,[t_{k-1}]}\right)\right] \\ &= \frac{3}{4}X_{\text{NDBI}}^{L,t_k} - \frac{1}{4}X_{\text{NDBI}}^{L,[t_{k-1}]} \end{aligned} \tag{15}$$

where the inner division by 2 normalizes the difference into the range $[-1, 1]$. In this formulation, both the single date and the differential contributions to the representation of the burned vegetation spectral signature are jointly exploited so that they can compensate for each other's variability across different geographical areas and seasons.

2) Burn Severity Index Threshold $\ell^{L,t}$: Although correlation exists between spectral measurements and fire severity, this functional relationship is strongly affected by ecosystem characteristics and seasonality of the geographical area under study [38]. Thus, any attempt to rely on global constants for large-scale applications is doomed to fail. To enhance and automatize the adaptation process, we rely on ecoregions[5] definition, being them ecosystems of regional extent [39]. In practice, through a trial-and-error process, an optimal threshold $\ell^{L,t}$ is found, which is dependent on the spatially dominant ecoregion over the spatial frame L. For simplicity, this choice is left independent from the temporal instant, i.e., $\ell^{L,t} = \ell^L$.

[5]https://ecoregions2017.appspot.com/

3) Spectral Index for Statistical-Based Screening X_{ssis}: As discussed in Section III-D2, weakly changed pixels (x, y) (i.e., those to be screened) are the ones behaving likely normal, where normality is estimated using past values from the temporal stack $T_{x,y}^{L,t_k}$. Considering that the statistical screening step is devised to eliminate sparse false detections (e.g., urban settlements and saturated pixels), the choice of a suitable index is made upon the need of reducing burned pixel values to anomalies w.r.t. the chosen spectral index. To this aim, indices that measure (to some extent) the vegetation content are all appropriate candidates. In particular, it turned out from some experimentation that such behavior is well catched by the CAI index. Therefore, we set $X_{\text{ssis}}^{L,t} = X_{\text{CAI}}^{L,t}$.

4) Abundant Threshold h for Full BA Reconstruction: The inclusive approach for BA reconstruction presented in Section III-D3 is based on the continuous nature of spectral indices as seen from a physical perspective. Within a specific spatial frame L, if candidate pixels have been found using threshold ℓ^L, spectral values of the burn severity index X_{bsi}^{L,t_k} which are below, but close, to the threshold are almost certainly related to the detected burned event. Since the threshold is chosen in a conservative way (i.e., it filters out most of the context by only passing seed pixels), after removal of weakly representative pixels (using the screening approach), reconstruction is done by aggregating pixels that are spectrally neighboring to the seeds. This spectral neighborhood is defined by the threshold $h > 0$, such that, given the physical meaning of the burned severity index, $0 < \ell^L - h$.

C. Morphological Parameters

1) Minimum Size of BA p: As stated in Section III-D2, parameter p determines the maximum size of what is considered residual spatial noise, right after the statistical-based screening. In principle, the value of p must be kept low in order to avoid degrading the intrinsic high spatial resolution of input data. However, due to adjacency or mixed pixel effects erroneous detections of size larger than 1 pixel can occur, thus we set as an empirical choice $p = 9$ pixels. This implies removal of components with area less than 0.36 ha in the Sentinel-2 case and less than 0.81 ha in the Landsat-8 case. Notice that, this choice precludes only the detection of BAs having extension of less than 1 ha, which is way less restrictive if compared to the commonly accepted size of a small fire, typically in the order of tens of hectars [14].

2) Minimum Inclusion Ratio i_R for Full BA Reconstruction: Following the inclusion principle for full BA reconstruction described in Section III-D3, a suitable choice for i_R would be any reasonable approximation of the lower bound on the ratio between the area of the fire seeds and the area of the abundant region including them, which is measured in real BA-related observations. This value can be seen as an intrinsic spatial–spectral characteristic of the land cover class involved by the fire event. In this work, the inclusion ratio lower bound is set empirically to $i_R = 0.03$.

V. SIMULATION OVER THE EFFIS AREA

In this section, the proposed BA detection system is tested in different locations within the EFFIS area and the quality

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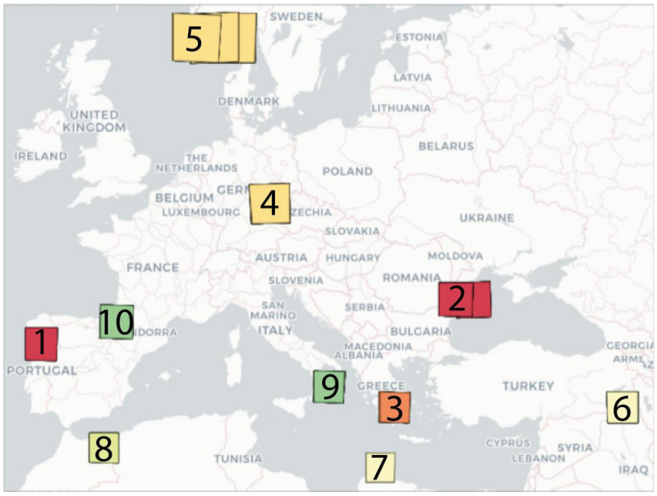


Fig. 2. Ten EFFIS locations over which the system simulation is performed.

TABLE IV

SENTINEL-2 DATASET. GRANULE AND ORBIT ARE DESCRIPTORS OF THE SPATIAL FRAME L. COLUMN n COUNTS THE NUMBER OF IMAGES IN THE BASE TIME SERIES

Country	Granule	Orbit	Sensing period	n
DEU	32UQA	022	13/02/2018 - 16/09/2018	39
DZA	30SXD	051	05/06/2018 - 29/08/2018	16
FRA	30TYN	051	26/01/2018 - 29/08/2018	42
GRC	34SFH	093	04/05/2018 - 01/09/2018	24
ITA	33SXD	036	05/05/2018 - 12/10/2018	32
LBY	34SEB	050	01/05/2018 - 30/06/2018	13
NOR	32VLL	051	01/04/2018 - 15/07/2018	22
PRT	29TNG	037	19/06/2018 - 22/09/2018	19
ROU	35TPL	007	04/03/2018 - 05/10/2018	43
TUR	38SKG	035	05/05/2018 - 27/09/2018	26

of the output is evaluated in comparative terms between the four spectral indices dNBR, dNBR2, dMIRBI, and wNDBI. In the following, details are given regarding the collection of data used in the simulations, the system implementation over a cloud-based infrastructure, the ground truth available, and the validation approach used to assess the quality of the BA maps produced.

A. Datasets

The dataset comprises images from year 2018 for ten different areas of interest (AOIs) in the EFFIS area (see Fig. 2). This is done both for Sentinel-2 and Landsat-8 data. The ten areas are chosen to maximize geographical and climatological diversity. The spatial frames and the reference sensing periods for the ten AOIs are listed in Table IV for the Sentinel-2 case and in Table V for the Landsat-8 case.

B. System Architecture and Implementation

The core module of the proposed BA detection system provides a bitemporal-based BA detection map for a given date t_k and a spatial frame L, the latter being a granule-orbit or path-row identifier in the case of Sentinel-2 or Landsat-8, respectively. This process is repeated for each image in

TABLE V

LANDSAT-8 DATASET. PATH AND ROW ARE DESCRIPTORS OF THE SPATIAL FRAME L. COLUMN n COUNTS THE NUMBER OF IMAGES IN THE BASE TIME SERIES

Country	Path	Row	Sensing period	n
DEU	193	025	12/01/2018 - 11/10/2018	16
DZA	199	036	14/05/2018 - 02/08/2018	6
FRA	200	030	29/01/2018 - 26/09/2018	15
GRC	183	034	06/01/2018 - 19/09/2018	17
ITA	187	033	10/05/2018 - 01/10/2018	10
LBY	183	037	11/03/2018 - 15/06/2108	7
NOR	199	019	12/04/2018 - 19/09/2018	11
PRT	204	031	20/07/2018 - 22/09/2018	5
ROU	181	029	08/01/2018 - 21/09/2018	16
TUR	170	034	01/04/2018 - 24/09/2018	12

the base time series on a spatial frame L. The collection of temporally subsequent BA detection maps provides a dynamic mapping of the evolution of the BAs in time. The system is implemented as an executable Python script that is able to run on AWS machines. One complete job consists of four main stages.

1) Get the spatial frame L and the target date t_k from an orchestrator.
2) Locally download the most recent m images with spatial frame L, from date t_k backward.
3) Compute BA detection map through processing.
4) Locally store the output BA map.

During the simulation, for each spatial frame listed in Section V-A, the corresponding base time series of n images is considered and a sequence of $n - m + 1$ jobs is launched to exhaust all possible target dates $t_m, t_{m+1}, \ldots, t_n$. The whole system scales up by parallelization of jobs executions over multiple AWS machines. For the simulations presented hereafter, an AWS `t2.2xlarge` instance is used, featuring 8-vCPUs and 32-GB RAM.

C. Reference Data and Validation Approach

The available reference information for validating the quality of BA maps is a collection of shapefiles manually drawn by photointerpreters and delineating the contours of fires occurred within each spatial frame L in the considered sensing periods. More specifically, each geometry in the shapefiles is conceptually obtained as an aggregation of all BA patches recognizable as changes from all possible image pairs in the images. Each geometry, therefore, represents the maximum extension of fire propagation and it has no specific temporal attribute. Reference delineation was aided by joint analysis of Sentinel-2, Landsat-8, and SPOT imagery.

This property of the validation dataset must be properly accounted for in designing the validation approach for the proposed system, which produces BA maps according to a sequential, nonincremental, mechanism. To understand this, an illustrative example is given in Fig. 3 where we observe that, because of occlusion generated by clouds, shadows, and smokes, there is no subsequent image pair from which the whole BA patch (contoured by red line) can be detected.

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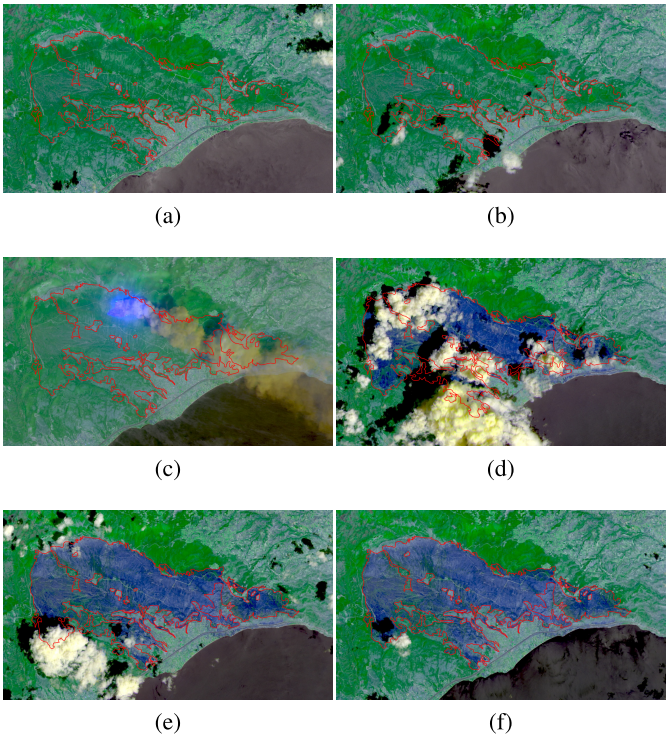


Fig. 3. Concept for delineation of BA reference geometries illustrated on a sequence of Sentinel-2 images from the GRC dataset. Images are displayed in false color assigning B04 (RED), B08 (NIR), and B12 (SWIR) to the red, green, and blue channels. The resulting reference geometry for the BA associated with this fire is superimposed to each image. (a) July 13, 2018. (b) July 18, 2018. (c) July 23, 2018. (d) July 28, 2018. (e) August 2, 2018. (f) August 7, 2018.

We account for this by designing a validation approach, which compares an aggregated BA detection map, obtained by pixelwise union of all BA maps, with the rasterized reference shapefile. The geometric accuracy of the detection is analyzed in terms of typical metrics that can be derived from the confusion matrix entries: true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). These metrics are F1 score (F1), overall accuracy (OA), CE, and OE. We also include an object-oriented evaluation metric, which compares the ratio between the number of detected connected components (N_d) having nonnull intersection with reference geometries and the number of reference connected components (H_r) to provide an indication of how many BA patches are hit in the detection. The considered metrics are therefore computed as

$$\begin{aligned} \text{F1} &= 2 * \text{TP}/(2 * \text{TP} + \text{FN} + \text{FP}) \\ \text{OA} &= (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \\ \text{OE} &= \text{FN}/(\text{FN} + \text{TP}) \\ \text{CE} &= \text{FP}/(\text{FP} + \text{TP}) \\ \text{HR} &= N_d/N_r. \end{aligned}$$

D. Optimal Thresholds

Following the criterion described in Section IV-B2, optimal thresholds ℓ^L are computed for all spatial frames L and

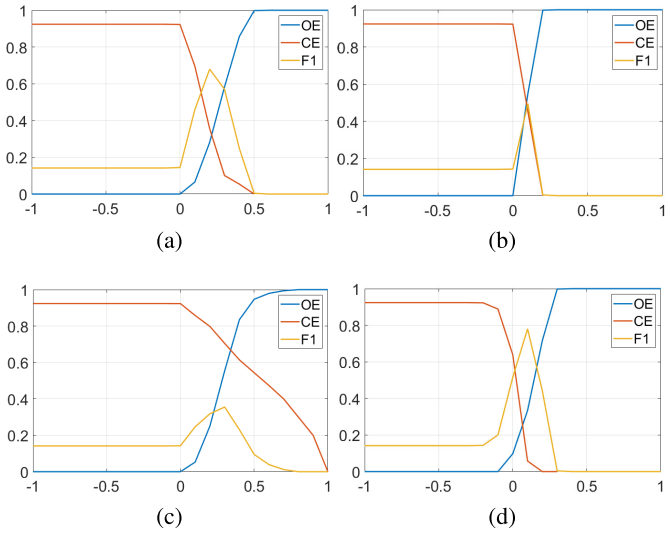


Fig. 4. Optimal thresholds selection for the GRC dataset, Sentinel-2 case. In each graph, for a grid of values ranging from −1 to 1, the curves of OE, CE, and F1 are plotted, for each spectral index considered. (a) dNBR. (b) dNBR2. (c) dMIRBI. (d) wNDBI.

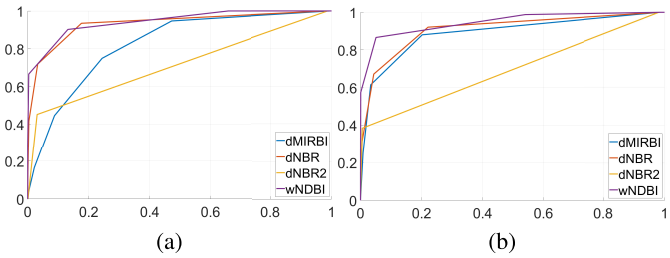


Fig. 5. ROC curves for the GRC dataset and both Sentinel-2 and Landsat-8 cases. (a) S2. (b) L8.

associated with their respective dominant ecoregions. In order to do this, the performance of BA detection is evaluated over a grid of values and the one that maximizes the F1 score is selected as the optimal. An example for the GRC dataset is shown in Fig. 4 where OE, CE, and F1 curves are plotted for the Sentinel-2 case and in Fig. 5 where ROC curves are plotted for both sensors. In the illustrated ROC graphs as well as in all the other AOIs, we report wNDBI performed better than the other indices, showing its consistency as a spectral index for binary decision.

E. Results and Discussion

The results of the validation of the proposed BA detection system are shown in Table VI for Sentinel-2 and Landsat-8 cases and the four considered indices. From the computational viewpoint, Sentinel-2 jobs took on average twice the time of Landsat-8 ones as the time series of images loaded is twice longer (ten images against 5). However, the overall time of computation of a job never gets above 10 min. We can observe looking at the metrics values in Table VI that the proposed BA detection method produces consistent results across sensors. This is a rather important fact that demonstrates how the spectral characteristics of BAs can be modeled and

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TABLE VI

ACCURACY METRICS EVALUATED ON THE FOUR BURN SEVERITY INDICES FOR BOTH SENSORS

		OE	CE	OA	F1	HR
Sentinel-2	dNBR	0.50	0.61	0.98	0.60	0.69
	dNBR2	0.62	0.66	0.97	0.41	0.60
	dMIRBI	0.55	0.71	0.96	0.33	0.55
	wNDBI	0.41	0.55	0.99	0.72	0.71
Landsat-8	dNBR	0.49	0.65	0.98	0.54	0.59
	dNBR2	0.60	0.70	0.97	0.39	0.55
	dMIRBI	0.63	0.59	0.97	0.37	0.56
	wNDBI	0.45	0.58	0.99	0.64	0.58

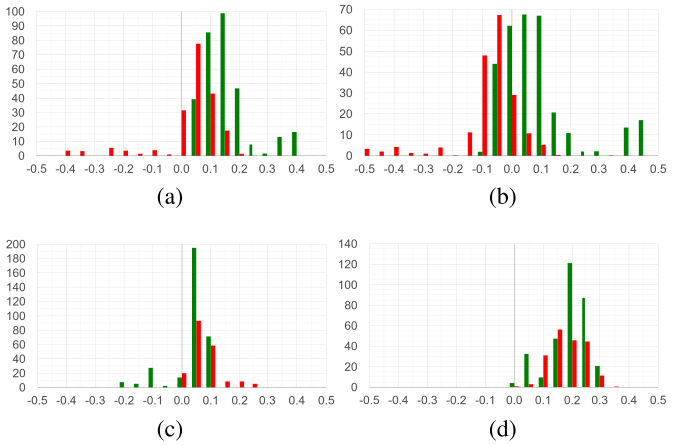


Fig. 6. Frequency histograms of spectral indices sampled both on burned spots (red bars) and managed crops (green bars) causing false alarms. First line shows single-time measurements: (a) NBR and (b) NDBI. Second line show bitemporal measurements: (c) dNBR and (d) wNDBI. Notice that using burn-oriented spectral indices, it is almost impossible to distinguish between BAs and crops.

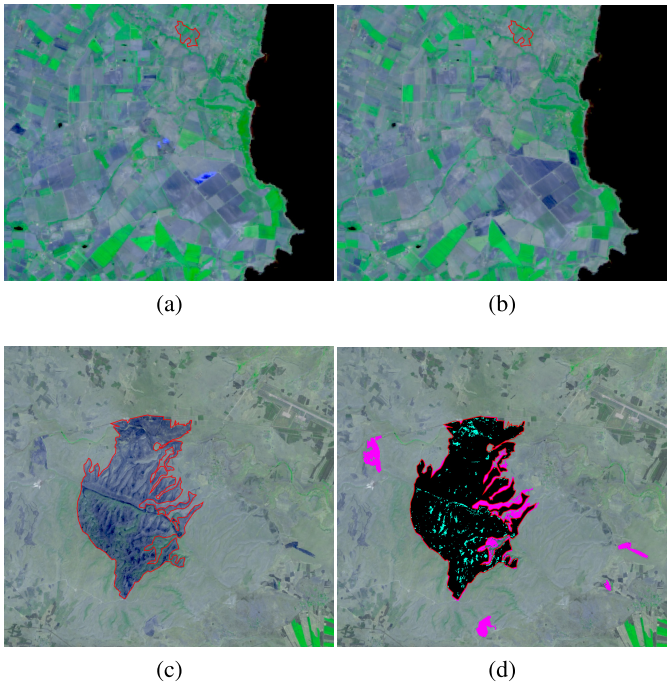


Fig. 7. Particulars from error analysis discussion. Images are displayed in false color assigning B04 (RED), B08 (NIR), and B12 (SWIR) to the red, green, and blue channels. Red lines are burned, which are shapes from reference data. First line shows a particular of a burning crop in ITA dataset at two dates: (a) July 4, 2018 and (b) July 14, 2018. Second line shows: (c) BA in TUR dataset (September 2, 2018) with (d) superimposed (September 2, 2018) the BA detection results (TP = black, FP = magenta, and FN = cyan), highlighting some pixels evaluated as missed alarms (cyan) because of overestimation of the BA in the reference.

exploited with a good degree of independence with respect to the sensor used.

The first fact that is clear from the table is that wNDBI outperforms the other three indices. In view of this, in the remainder of the section, the results related to the proposed wNDBI are those discussed. Even if, in absolute terms, the measured error rates are slightly worse than those in the state of the art (cf. Section II-D), we must consider some crucial facts: 1) the proposed algorithmical flow is simpler than those proposed in the literature, as our main goal was to assess the performance of spectral index thresholding and mitigation techniques to improve the detection; 2) the BA detection task is accomplished here with no supervision and massively on every date with available imagery, posing a serious challenge as no prior knowledge is assumed on fire locations and dates of events, neglecting the possibility for a supervised choice of optimal image pairs for the detection; and 3) our reference is generated manually and, therefore, more precise than other reference obtained via supervised classification [13].

From the average quality of the results presented in Table VI, see also some examples of detection/reference pixelwise comparisons shown in Fig. 8, we can see that BA maps can be obtained from simple index thresholding, provided that mitigation techniques are applied to undertake typical error patterns. Indeed, two main sources of errors are identified in our experiments, impacting in different ways on the overall quality of the results. The first one is cloud cover, which is not critical to a large extent, and the second one is the managed crops, being this one more critical source of false alarms in some circumstances.

The presence of a cloud determines a lack of information on a specific date, but its impact is different depending on how the thematic classification represents it. In the case a cloud is missed in the thematic map, the corresponding pixels are processed and the resulting errors are essentially random. It is empirically observed that those errors are mainly identified as false alarms clustered around cloud crowns, where the measured reflectance rapidly fluctuates between very low and high values in all bands. In the case clouds are correctly masked, we avoid the random interference, but we are left with the lacking information from occluded pixels. Of course, for cloudy pixels in the target date, no detection can be done and only future observations could mitigate for associated errors. On the other hand, for a burned pixel in the target date, the availability of a temporally close vegetated nonoccluded pixel would result in detection. Temporal persistence of clouds on the same spot would inevitably result in missed alarms. However, if the cloud cover is only partial (in spatial terms), BAs can be reconstructed at least partially, resulting

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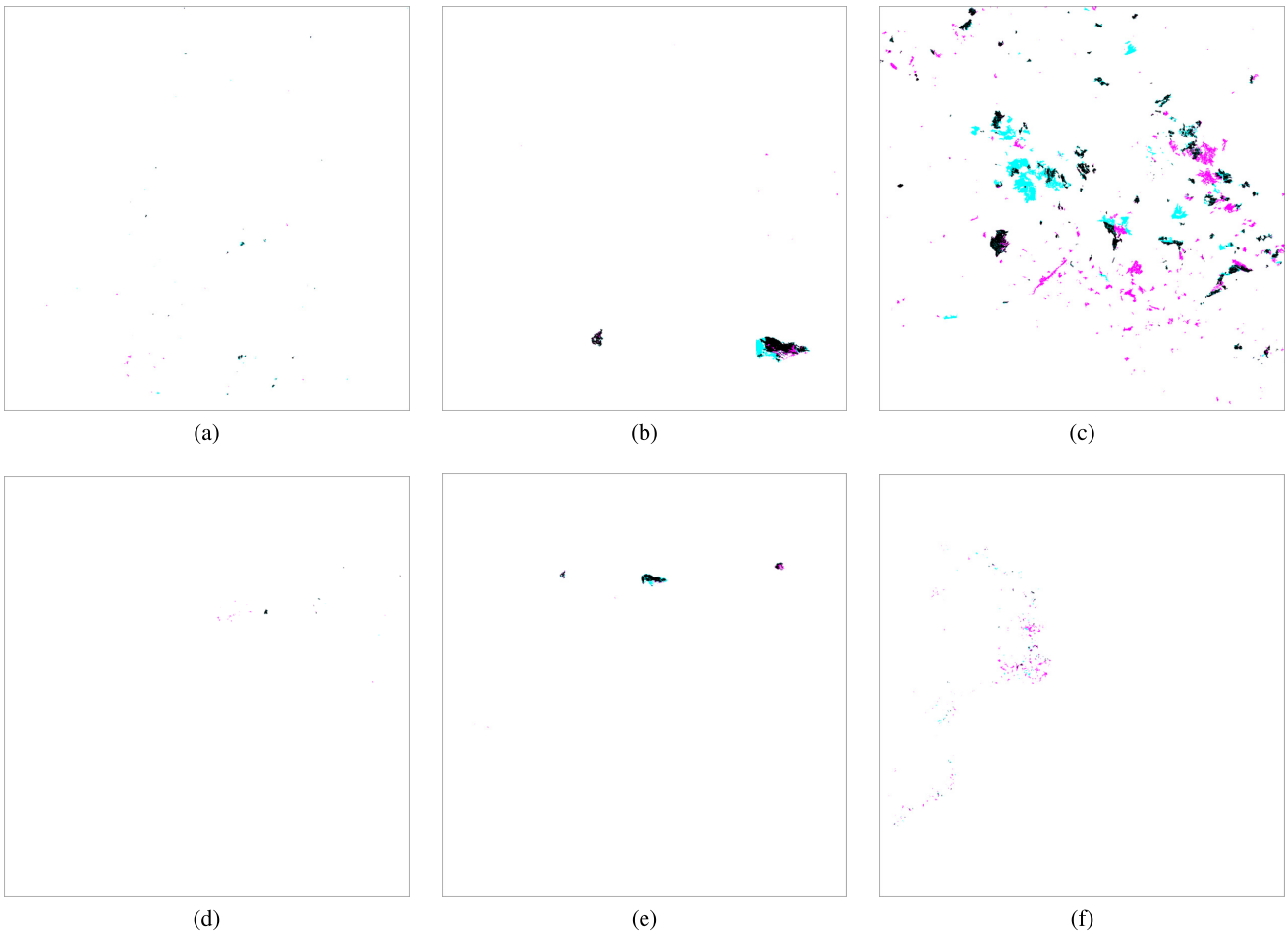


Fig. 8. Plots of aggregated BA detection maps compared with reference maps for both sensors, three AOIs each. First line shows three Sentinel-2 outcomes: (a) PRT, (b) GRC, and (c) TUR. Second line shows three Landsat-8 outcomes: (d) DZA, (e) GRC, and (f) ITA. Legend of colors is: TP = black, TN = white, FP = magenta, and FN = cyan.

in high hit rates with possibly nonnegligible missed alarm rates. Explanatory is the GRC case (Attica fires), as shown in Fig. 8(b) and (e), where all BAs are hit (HR = 1.00), but the entire patch cannot be reconstructed because of partial cloud cover in all observations (cf. Fig. 3); indeed, OE is 0.31 and 0.25 in the Sentinel-2 and Landsat-8 cases, respectively. Most of the other cases with high missing rates behave similarly, being compensated by effective high hit rates indeed. An edge case of the cloud cover issue is the FRA dataset (Pyrenees fires), where almost every image is heavily cloudy and the few identifiable fire spots have in many cases the corresponding pixel in the preimage that is covered with snow. In some of these cases, BAs are missed.

False alarms follow in general the trends already seen in many works dealing with BAs and/or AF detection [13], [21], [30]. However, the large extension of the study dataset allows us to better understand the spatial and temporal distribution of such errors, which are not ubiquitous but very dependent on certain conditions. Essentially, false alarms can be characterized by typology of the land cover class being erroneously detected as burned. The one class that dominates false alarms typology is crops, specifically, not all crops but only those undergoing certain agricultural seasonal practices. The magnitude of the problem for some AOIs, such as ITA, FRA, and DEU, is not negligible, reaching CE greater than 0.40. However, we must consider that the last step of the processing chain aggregates all pixels from a spectrally homogenous patch starting from some seeds. Therefore, even if candidate pixels erroneously detected in a crop are just a few, the full crop is detected as burned whenever the inclusion ratio threshold i_R is crossed. As a matter of fact, the threshold allows for eliminating very large crops, but it is not sufficient for complete removal of medium-to-small-sized ones.

The discrimination between burned vegetation and crops in specific seasonal moments is a very difficult task, even impossible through the usually burned index approaches. This is clearly represented in Fig. 6, where the spectral response of pixels sampled both on burned spots and managed crops causing false alarms is plotted in histograms. We can observe that there exists no effective way to discriminate between the two classes via thresholding approach at the spectral/spatial resolution of Sentinel-2 and Landsat-8. It is worth to be noted that this difficulty is also relevant in the context of manual drawing of reference geometries of the BAs, especially when auxiliary higher resolution data are not available to support decision

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for the photointerpreters. The interpretation of crop-related errors is further worsened when the observed area presents cases of crop residue burning due to certain practices in crop management. An example is given in Fig. 7(a) and (b), where a fire is expanding in a crop area (the fire front is clearly visible) in the first image. By missing the particular of the AF (this may happen for several reasons, e.g., cloud cover and no image available), it is very difficult to reference the crop as burned solely relying in the second image. Indeed, this burning crop is not referenced as BA, while the detector gets it. Another important consideration relates to the temporal distribution of these false alarms and its impact on the validation results that are obtained via temporal aggregation of all BA detection maps. The critical cases of heavy false alarms are very sparse in time, leading us to believe that the matching condition for the verification of false detection is probably a seasonal event in the cultivation cycle of the crops. Notice that in the aggregative validation process of our simulations, for each AOI measuring high CE, at most one or two BA maps contributed for more than 99% of the total amount of false detections. In conclusion, interpretation of BA detection results over crop areas must be carried by considering all the above-mentioned aspects.

To conclude, an almost negligible but worth mentioning source of missed alarms is in general due to the manual drawing of reference geometries by photointerpreters. It seems that the human eye tends to slightly overestimate the extention of the fire being unable to identify possible unburned spots within its perimeter at the given resolution of 20 and 30 m. In fact, some sparse missing alarms patches can be noticed within the boundaries of detected BAs in many of the AOIs that correspond to areas of green vegetation in the post image as seen from false-color renditions, as shown in one clear example in Fig. 7(c) and (d).

VI. Conclusion

Modern technologies, such as satellite-borne optical sensors and powerful computing infrastructures, are in some cases made available at no cost (e.g., GEE) to a broad audience, helping in making global change information available to more and more people. BAs have become a major subject of study because of their strong link to climate change, with its dramatic and recognizable impacts in societal dynamics. Moles of freely available MS data coming from Sentinel-2 and Landsat-8 missions are vital for the generation of BA reference datasets and/or operational products.

Most of the times BA information can be recovered from dedicated spectral indices obtained from relatively simple band algebra. However, simple approaches can aid BA estimation only if typical error patterns are known and accounted for, especially when working at large (e.g., continental) scales. This article proposes an automatic BA detection system based on burn severity index thresholding, which integrates dedicated false and missed alarm mitigation strategies to improve the detection accuracy. In the proposed framework, we demonstrated that simple processing steps implemented to accomplish burned index thresholding can reach accuracy bounds slightly larger than the state of the art, which in some cases can be accepted as a tradeoff between more accuracy at the cost of impractical computational complexity. In this framework, we produced results on an extended simulation over the EFFIS area. As an outcome, a detailed analysis on the quality of the results brought us to the conclusion that the errors degrading the performance are not random, but they have a rather clear characterization, with a clear predominance of false alarms related to some stage of crops management. Although the recorded errors have a nonnegligible impact on the overall performance, we see that they are essentially well characterized and recognizable. This fact can be exploited constructively to devise *ad hoc* strategies for improving the outcome of the detection by focusing on the specific class of crops. To this aim, our experience tells that efforts in mitigating detection errors operating at the level of burn severity indices are disadvantageous in nature: actions taken to contain such errors have very strong overall impact even in noncritical areas, potentially worsening the collective performance.

In this respect, some concluding remarks are needed. If improved accuracy of the thematic products associated with MS imagery would be surely beneficial for containment of certain types of erroneous detections, especially missed spots due to cloud cover, also adoption of different temporal pairing strategies is a valuable option. For example, information carried in several pairings of the type (t_{k-h}, t_k), for some $h > 1$, is certainly more complete than that available from (t_{k-1}, t_k). As already discussed, the major threat to a successful mapping of BAs through spectral index analysis is represented by false alarms in corresponding to some stages of crops management. As a direct consequence of our discussion of the results, we see that filtering out crops from the analysis (a task that is anyhow not straightforward) will not represent the best choice at all. The main reason for this is that often BAs are found within crops (burned cultivation residuals) and this approach would prevent them from being detected. Therefore, more involved techniques need to be devised to cope with this category of false alarms.

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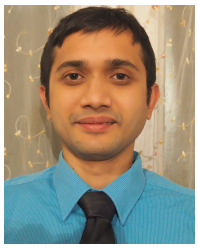


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